

Predictive Attrition Risk & Workforce Retention Strategy

GCI Global 2025- FINAL ASSIGNMENT

Agenda

- **Introduction**
- **Background & Objectives**
- **Data & Approach**
- **Key Findings**
- **Impact**
- **Recommendations**



ATTRITION

EXECUTIVE SUMMARY

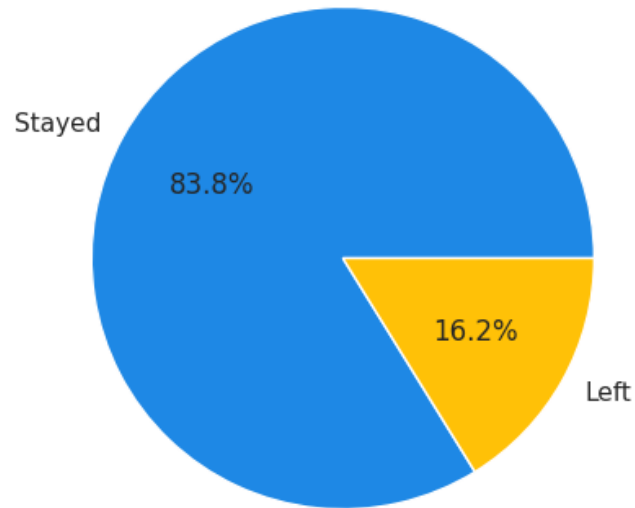


Employee attrition is a critical global challenge, with benchmarks showing 18–22% in India (Naukri JobSpeak, 2023) and 17.4% across European tech firms (Ravio, 2025), ranging from 42.7% in retail to 16.4% in financial services.

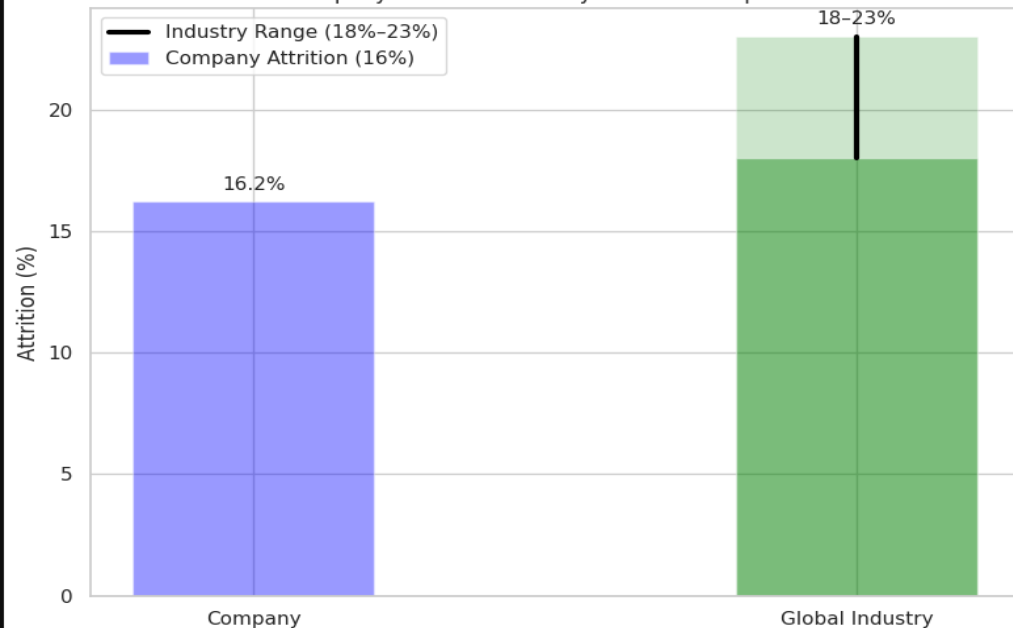


This project develops a predictive attrition model integrating HR, performance, compensation, stress, and work-life data. Two best-performing models—Logistic Regression (88.78% accuracy, 58.23% F1) and XGBoost (89.46% accuracy, 56.34% F1)—with SHAP explainability identify top-risk employees for targeted retention interventions, delivering data-driven strategy backed by financial impact modeling (SHRM, 2022).

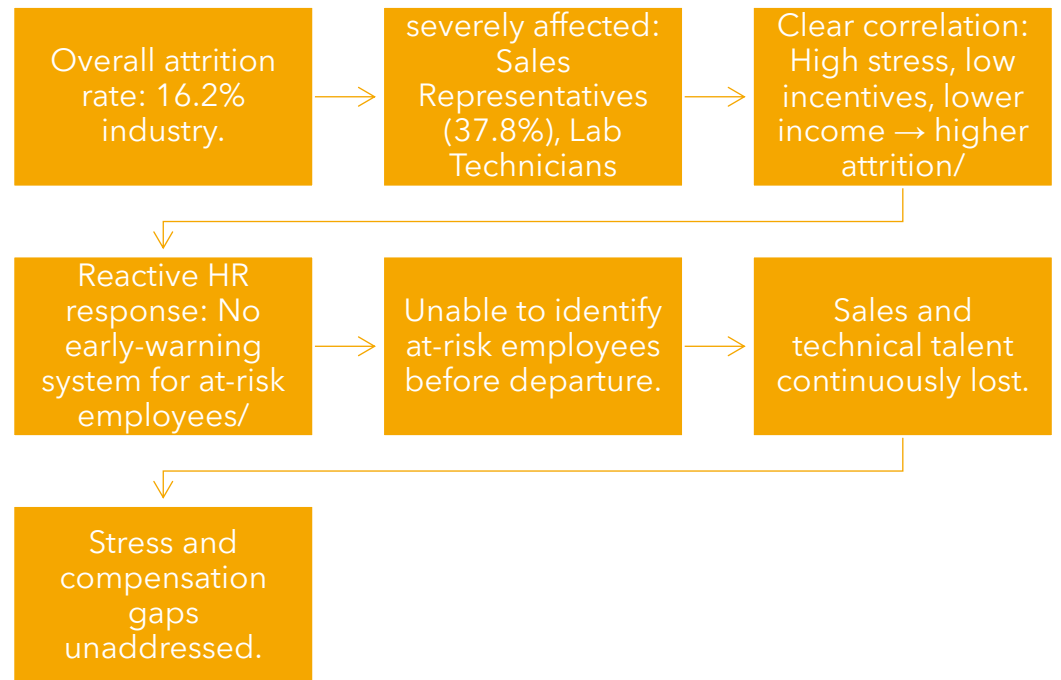
Attrition Distribution



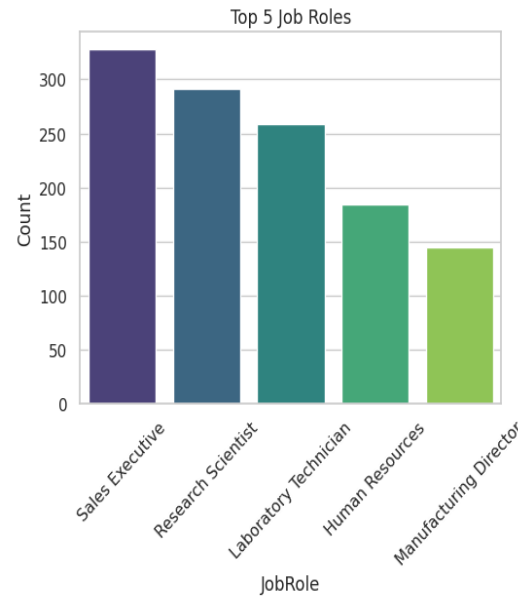
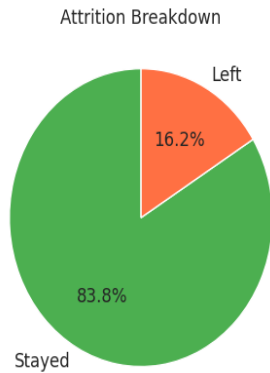
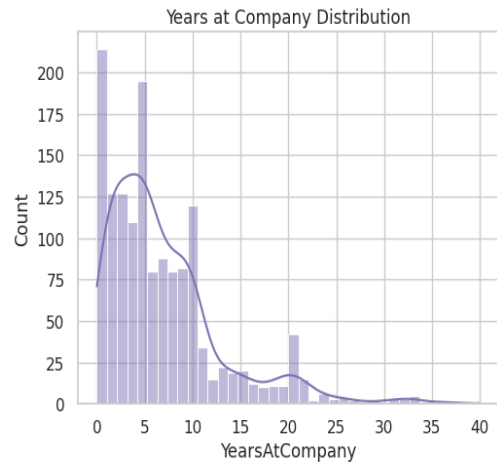
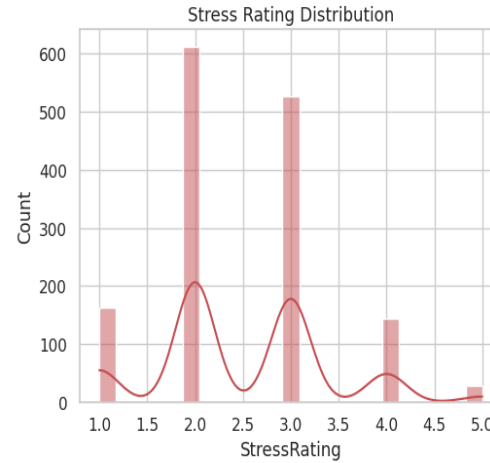
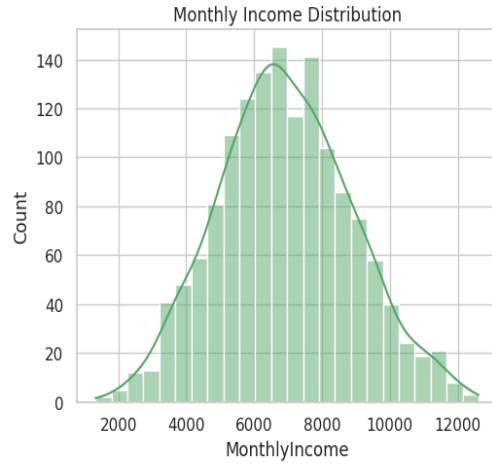
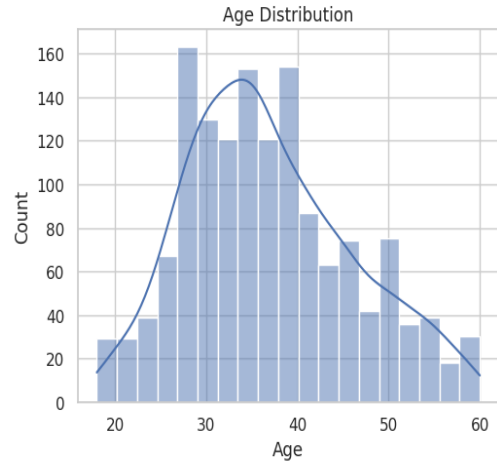
Company vs Global Industry Attrition Comparison



BUSINESS PROBLEM



DATA OVERVIEW



1,470 complete
employee
records,

44 baseline
features
analyzed,

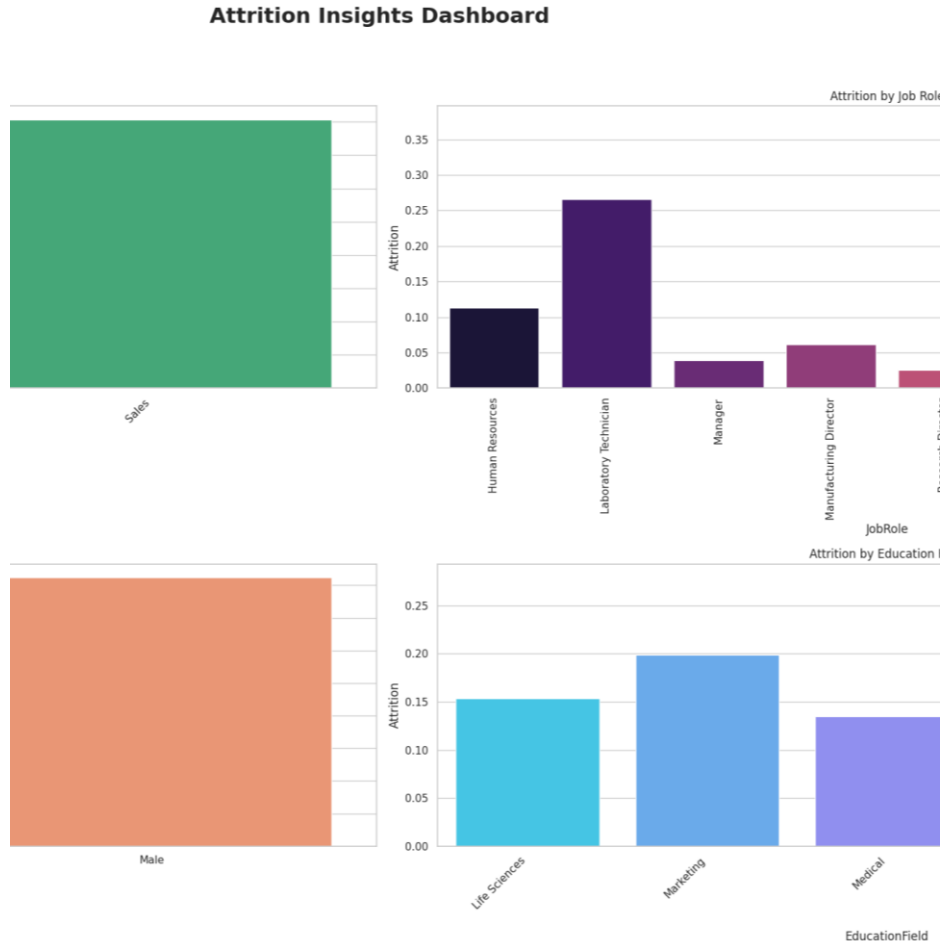
12+ engineered
features
created,

Zero critical
missing values,

5+ years of
historical data,

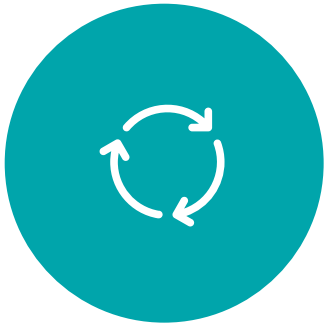
Class imbalance
corrected

Exploratory Data Analysis: Attrition Patterns



- Sales department and sales representative roles show the highest attrition at 20% and 37% respectively ie nearly 3x the company baseline.
- Interestingly, attrition is equal across genders (~16%), suggesting systemic workplace factors (stress, compensation, workload) rather than demographic biases.
- Employees with technical educational backgrounds retain better (13% attrition) vs. medical/non-technical backgrounds (20-26%)—indicating potential skill-mismatch or role-satisfaction differences.
- These patterns are not random—they're predictable and actionable. Our ML model will identify which individuals within high-risk groups are most likely to leave, enabling targeted, cost-effective interventions.

ML-Driven Attrition Detection



Models Evaluated:

Logistic Regression, Random Forest, XGBoost, Gradient Boosting, LightGBM, CatBoost, SVM, KNN



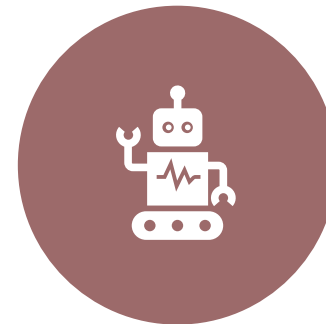
Pipeline:

EDA → Feature Engineering → Model Benchmarking → Explainability (SHAP) → Threshold Tuning → Business Interpretation



Evaluation Focus:

- F1 Score (Attrition Class)
- Precision, Recall
- AUC
- Confusion Matrix
- Calibration



Model Rationale:

•8-model framework spans three layers: Interpretability (Logistic Regression + Random Forest) decodes attrition drivers through coefficients and feature importance; Performance (XGBoost + LightGBM + CatBoost) maximizes accuracy via gradient boosting; Validation (Gradient Boosting + SVM + KNN) stress-tests robustness. This balances stakeholder transparency with production-grade predictive power, answering both *why* and *who* will leave.

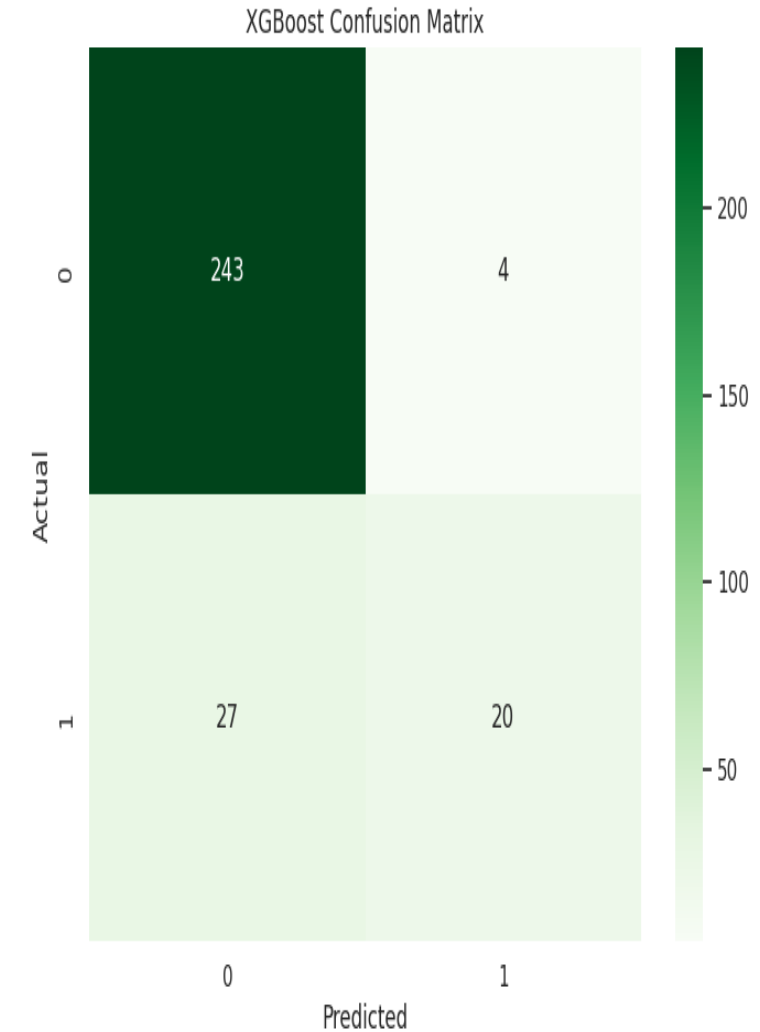
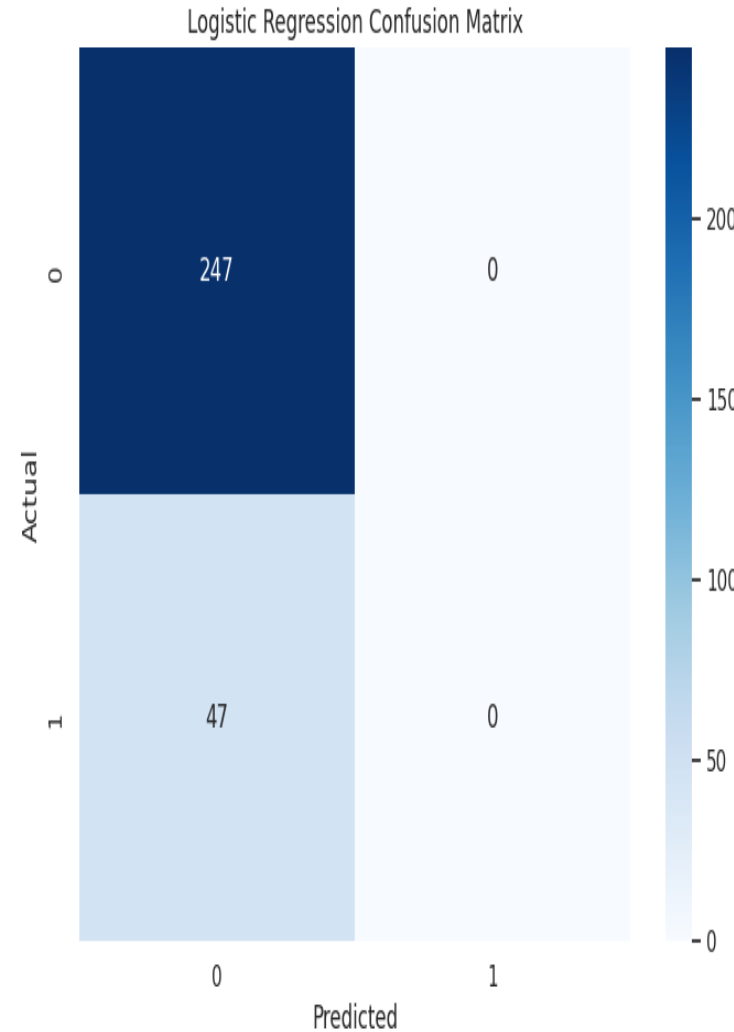


Model Benchmarking

Model	Accuracy	Precision(1)	Recall(1)	F1(1)
XGBoost	0.8946	0.83	0.43	0.5634
Logistic Regression	0.8878	0.72	0.49	0.5823
Gradient Boosting	0.8844	0.76	0.40	0.5278
CatBoost	0.8844	0.84	0.34	0.4848
Random Forest	0.8639	0.82	0.19	0.3103
SVM	0.8401	0.00	0.00	0.0000
KNN	0.8333	0.43	0.13	0.1967

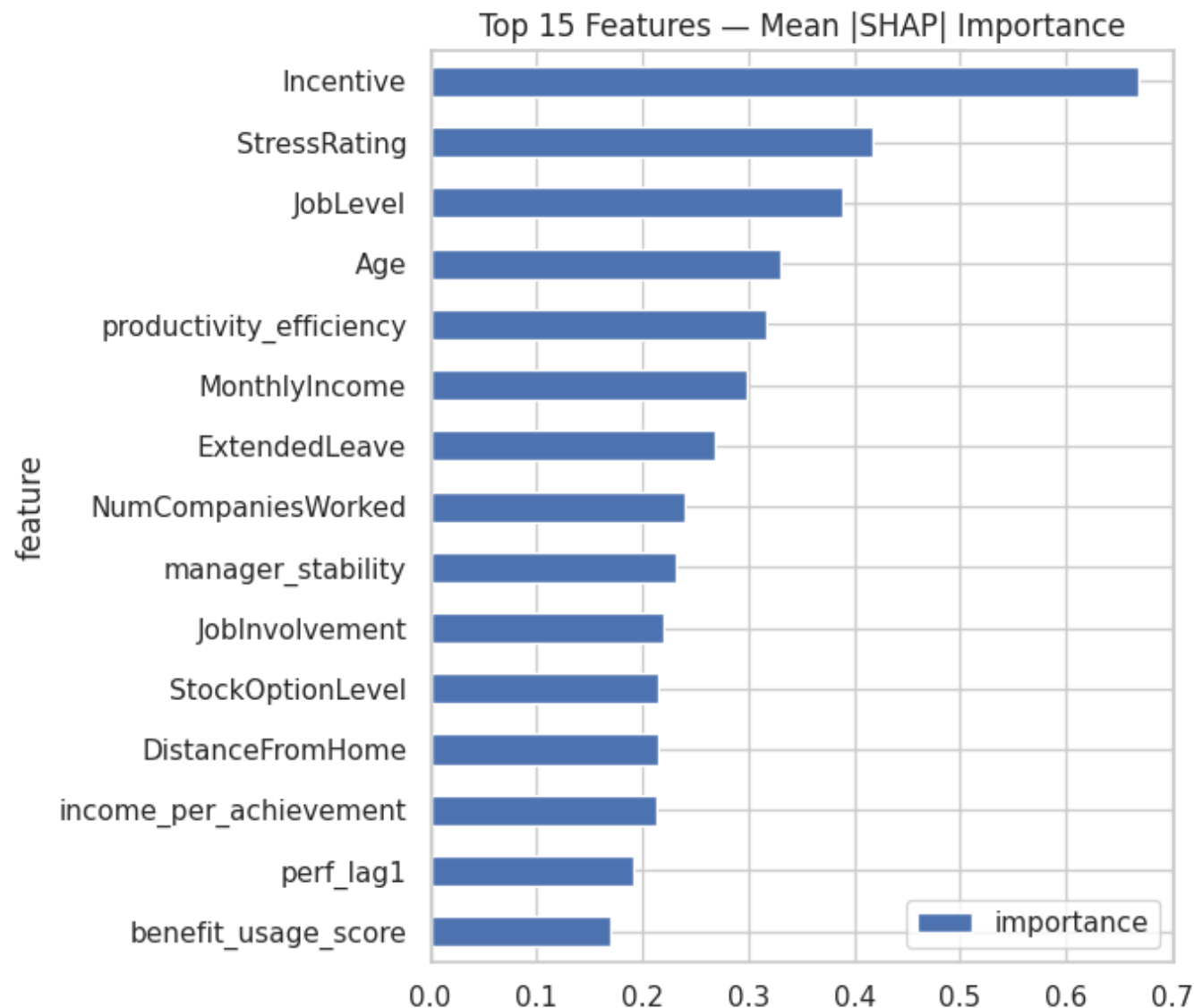
CONFUSION MATRIX

- Although Logistic Regression delivers high overall accuracy (88.8%) and a reasonable F1 score (0.58), the confusion matrix demonstrates a critical limitation: it fails to capture any true attrition cases, consistently predicting retention only. This apparent performance is inflated by class imbalance, as non-attrition cases dominate.
- For business impact, reliably identifying at-risk employees is paramount—making class-specific metrics (precision, recall) and confusion matrix analysis essential. XGBoost, with its higher recall for attrition, is better suited for targeted HR intervention, despite a slight drop in overall accuracy.



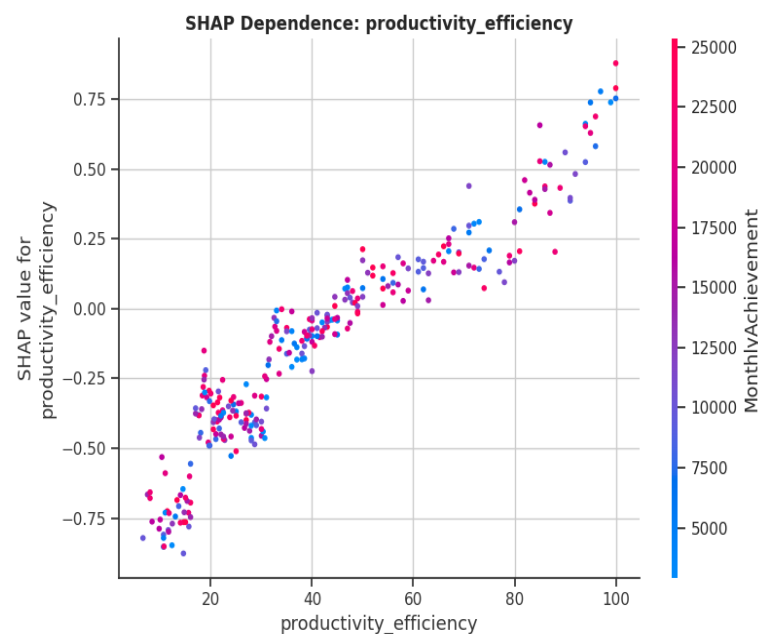
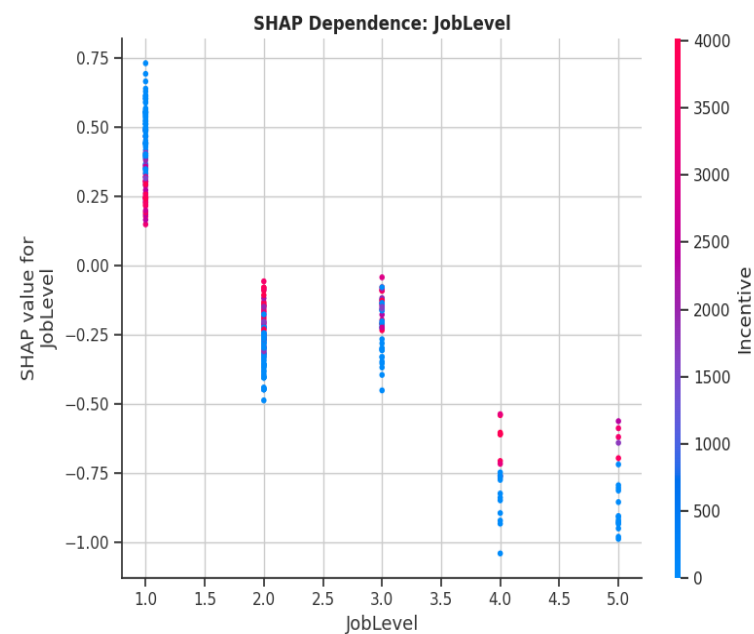
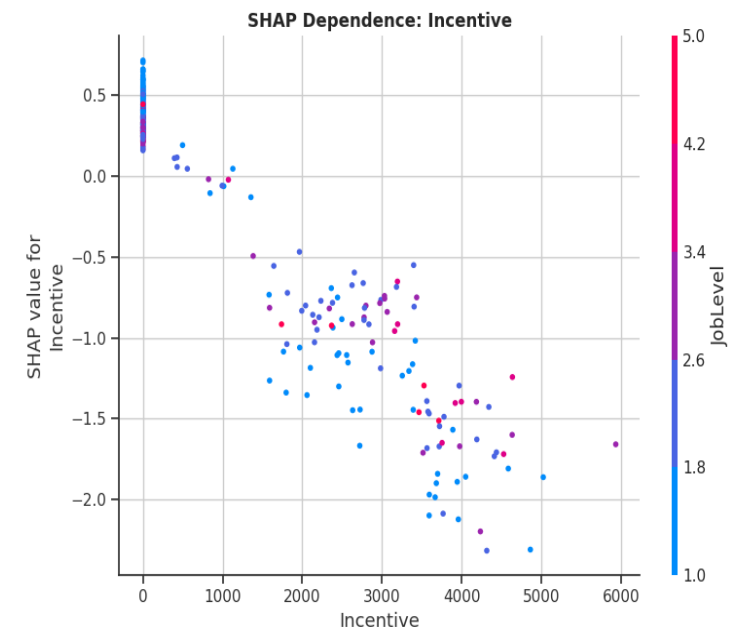
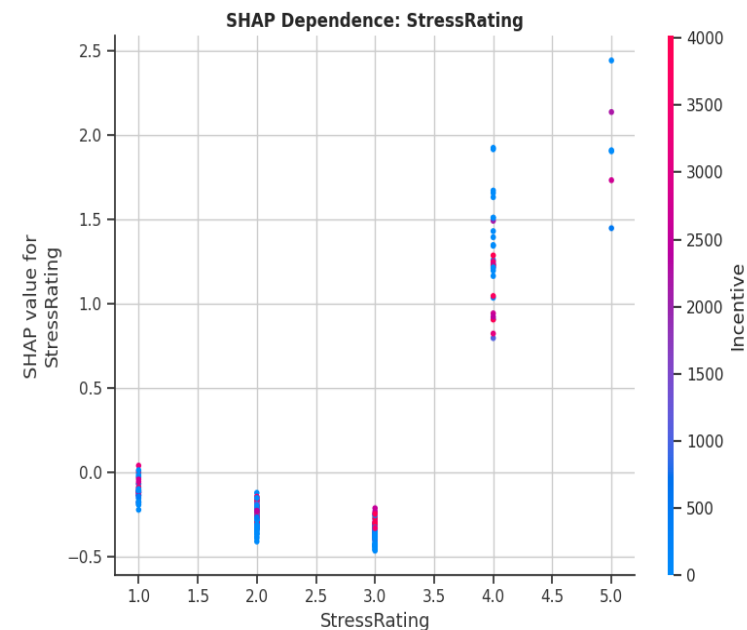
Feature Importance & SHAP Summary

- Top 5 Attrition Drivers Identified:
- Incentive (0.67) – Strongest predictor; lack of incentives dramatically increases attrition risk
- Stress Rating (0.42) – High workplace stress is the second major driver
- Job Level (0.40) – Career progression and seniority influence retention
- Age (0.33) – Younger employees show higher flight risk
- Productivity Efficiency (0.32) – Low productivity correlates with higher attrition

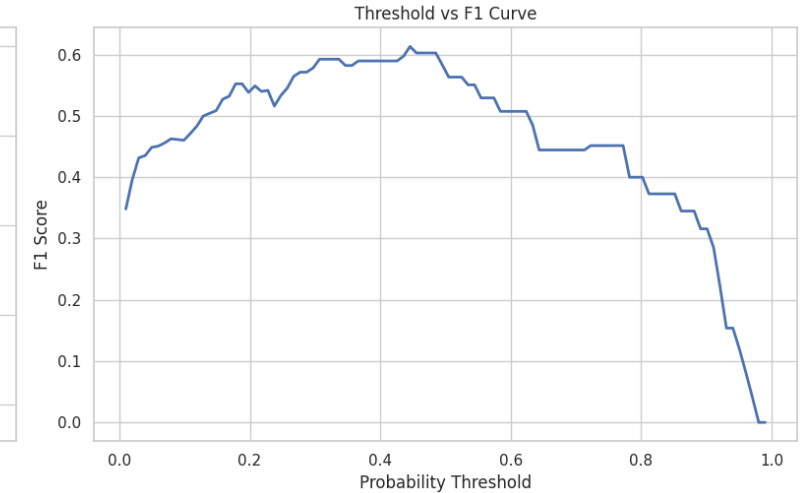
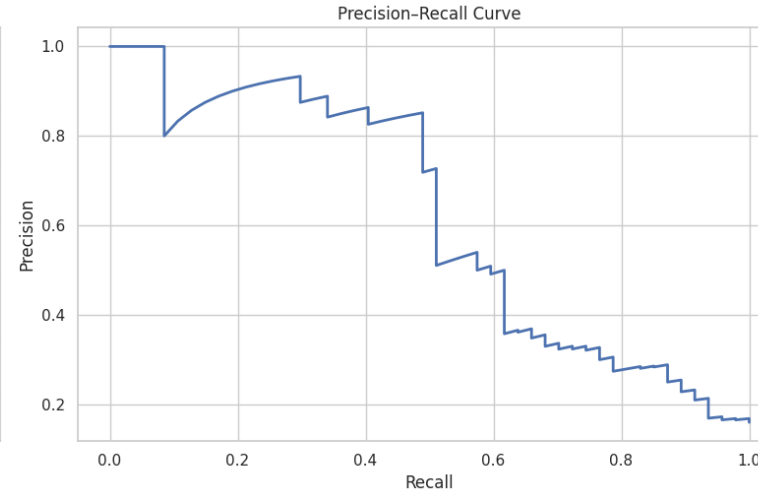
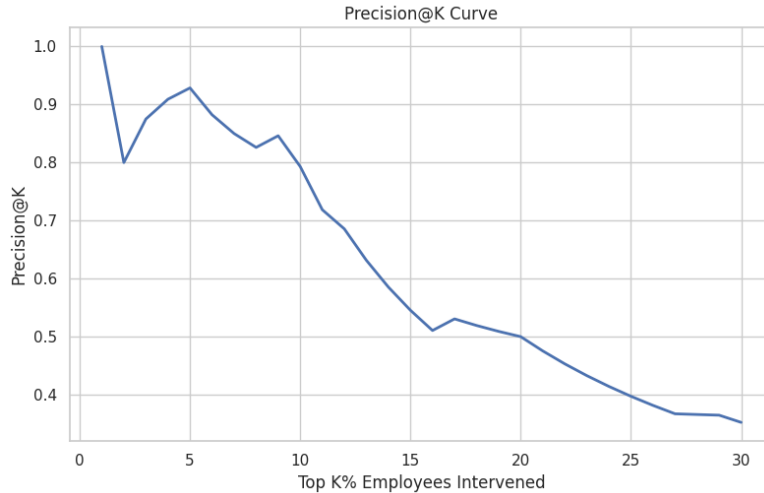


SHAP Dependence Plots

- StressRating (Top-Left):
 - SHAP values rise sharply at stress level 4 and especially 5—high stress is a strong predictor of attrition.
 - Lower incentives (color bar) cluster at high stress levels, amplifying risk.
- Incentive (Top-Right):
 - Attrition risk drops as incentives increase; most attrition occurs below ₹2000 incentive.
 - JobLevel coloring shows the negative impact is strongest for junior roles (blue).
- JobLevel (Bottom-Left):
 - Higher job levels (4, 5) are associated with lower SHAP values, indicating lower attrition risk.
 - Retention is strongest when incentives are higher (pink).
- Productivity Efficiency (Bottom-Right):
 - SHAP values increase with higher productivity efficiency—high performers are less likely to leave.
 - MonthlyAchievement color bar shows the retention effect is greatest for those with higher achievements.



Model Evaluation Metrics



Threshold Tuning & Precision@K

Precision@K Curve (Left):

- Highest precision for the top K% at-risk employees, exceeding 90% when targeting the top 5% most likely to leave.
- Precision decreases as broader populations are targeted, but remains above 70% up to the top 10%—supporting targeted retention interventions.

Precision-Recall Curve (Middle):

- The model maintains high precision (>80%) even as recall increases, showing effective identification of true positives with minimal false positives.
- Precision drops sharply at higher recall, consistent with challenging class imbalance in attrition prediction.

Threshold vs F1 Curve (Right):

- Maximum F1 score is achieved at a probability threshold around 0.4-0.5.
- Threshold tuning helps balance sensitivity (recall) and specificity (precision) for optimal attrition detection.



High-Risk Employee Identification

Rank	Employee ID	Attrition Risk	Stress Level	Job Role	Department
1	1944	97.5%	4	Human Resources	HR
2	19	96.5%	2	Laboratory Technician	R&D
3	282	95.8%	4	Healthcare Representative	R&D
4	1464	94.8%	5	Laboratory Technician	R&D
5	1551	94.7%	2	Laboratory Technician	R&D
6	1487	92.6%	4	Sales Executive	Sales
7	1649	92.1%	5	Laboratory Technician	R&D
8	1079	91.9%	3	Laboratory Technician	R&D
9	137	91.2%	1	Laboratory Technician	R&D
10	190	90.4%	3	Laboratory Technician	R&D

Initiative	Target	Expected Impact	Timeline
Incentive Redesign	Technical roles + Sales	12-20% attrition reduction	Immediate
Stress Reduction Programs	High-stress teams (Stress 4-5)	15-25% retention gain	30-60 days
Career Pathing	Mid-level roles (3-5 yrs tenure)	Combats mid-career exodus	Q1-Q2
ML-Powered HR Dashboard	HR Management	Real-time early warnings	Ongoing

Market Context & Strategic Recommendations

